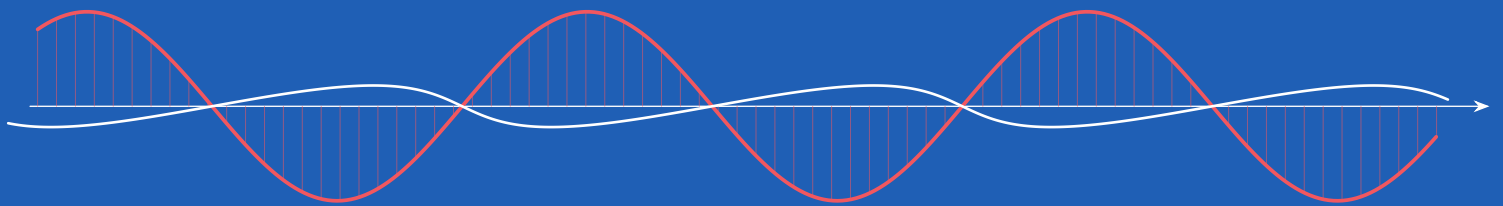


Electromagnetics

الکترومغناطیس



Ferdowsi University of Mashhad

دانشگاه فردوسی مشهد

Instructor: سید محمد سعید ماجدی

Compiled and edited by: Aidin Shekari

Electromagnetics

الکترومغناطیس

Lecture Notes

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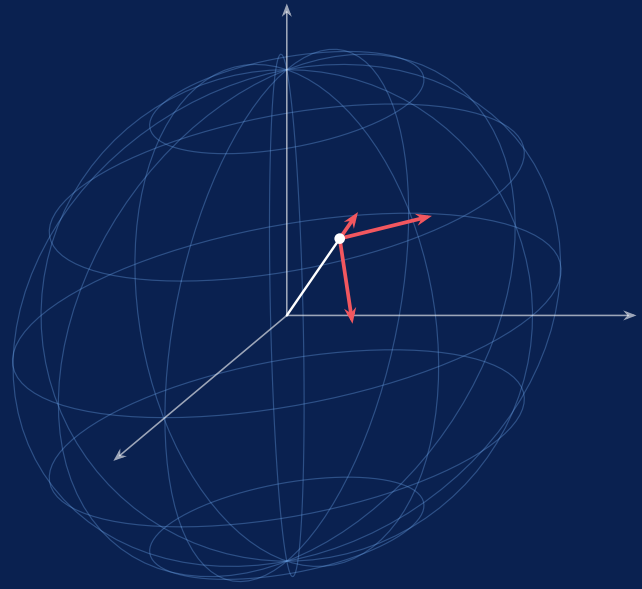
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دانشگاه فردوسی مشهد

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1



CHAPTER

Design Specimen

نمونه طراحی

In this chapter

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1.1 About this specimen

This chapter is a typesetting specimen of the design system. Its sentences and formulas are placeholders chosen only to exercise the layout; they are not course content and are never included in the book.

Inline mathematics in running text: $x, y, z, r, \theta, \phi, \rho, \mathbf{E}, \mathbf{H}, \mathbf{D}, \mathbf{B}, \mathbf{J}, \sin \theta, \cos \phi, \ln r, e^{j\omega t}, \operatorname{Re}\{z\}, \operatorname{Im}\{z\}, \frac{1}{x}$ and $\mathbf{a}_x$.

► Displayed formulas

$$\begin{aligned}\mathbf{A} &= A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z, & |\mathbf{A}| &= \sqrt{A_x^2 + A_y^2 + A_z^2} \\ \nabla f &= \frac{\partial f}{\partial x} \mathbf{a}_x + \frac{\partial f}{\partial y} \mathbf{a}_y + \frac{\partial f}{\partial z} \mathbf{a}_z, & \nabla \cdot \mathbf{A}, & \quad \nabla \times \mathbf{A}, & \quad \nabla^2 f \\ \oint_C \mathbf{A} \cdot d\mathbf{l}, & \quad \oiint_S \mathbf{A} \cdot d\mathbf{s}, & \quad \iiint_V f \, dv, & \quad \sum_{n=1}^{\infty} \frac{1}{n^2}, & \quad \lim_{x \rightarrow 0} \frac{\sin x}{x}\end{aligned}$$

1.2 Blocks

≡ Definition | Placeholder title

Body text of a definition block, with an inline formula $f(x) = \frac{1}{x}$.

$$g(x) = \int_0^x f(t) \, dt$$

▽ Theorem

Body text of a theorem block.

└─ Proof.

Body text of a proof block; the square at the end marks its close.

$$a = b \quad \Rightarrow \quad a + c = b + c$$

■

∧ Lemma

Body text of a lemma block.

⊙ Proposition

Body text of a proposition block.

→ Corollary

Body text of a corollary block.

⚡ Important result

Body text of an important-result block.

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

🔗 Example

Body text of an example block.

➤ Solution.

Body text of a solution inside the example.

💎 Exercise

Body text of an exercise block.

📌 Remark

Body text of a remark block.

1.3 Figures and tables

A paragraph that introduces the figure below.

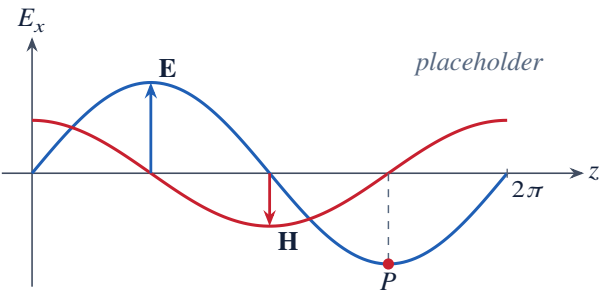


Figure 1.1 | A placeholder figure: axes, a curve, vectors and labels

Quantity	Symbol	Unit
placeholder A	E	V/m
placeholder B	H	A/m
placeholder C	ρ	C/m ³

- First item of a list.
- Second item of a list, with θ .